



Parktown Boys' High School

Geography Department

Grade 12 Mapwork Test

Examiner: Mr. D Grace
Moderator: Mr. A Meintjes

Time: 1 Hour
Total: 60 Marks

Instructions:

1. Make sure you put your **name** and **class teacher** on the top of your answer sheet
2. Answer **ALL** the questions on your answer sheet
3. Number the answers correctly according to the numbering system used in this question paper.
4. All diagrams must be at least 7 – 10 lines in size and fully labelled
5. Write neatly and legibly
6. GOOD LUCK!

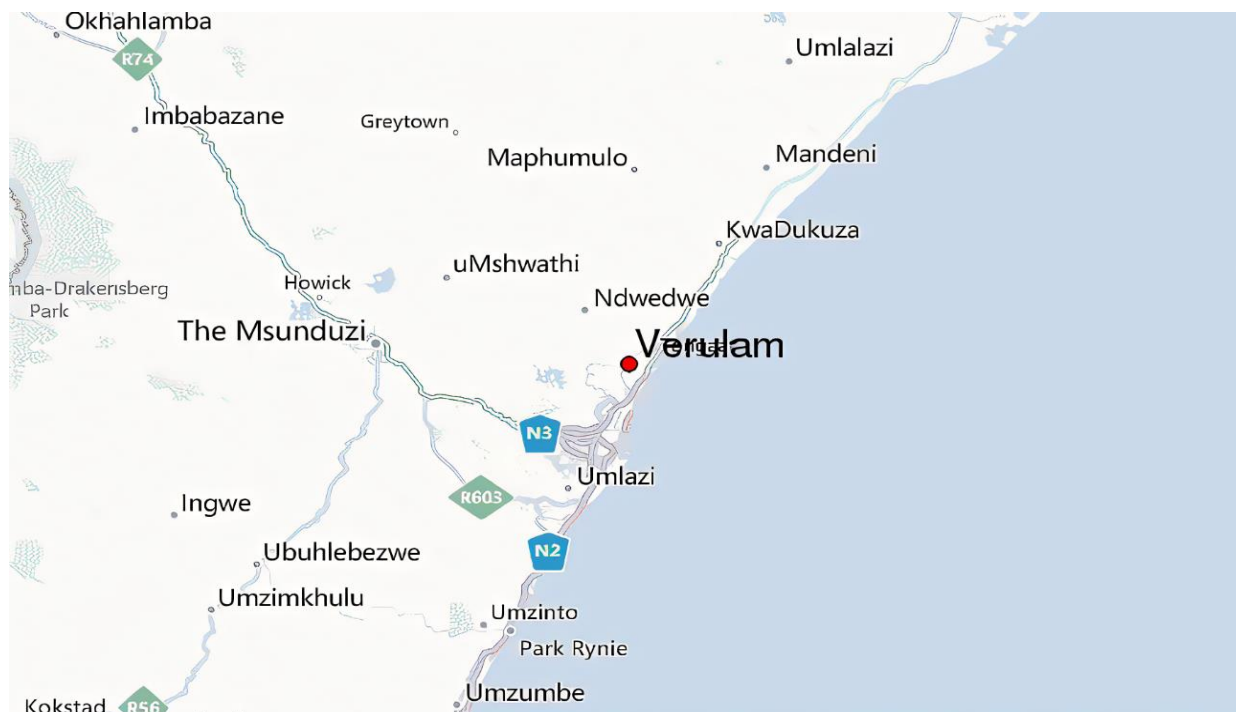
RESOURCE MATERIAL

1. An extract from the topographic map 2931 CA VERULAM
2. Orthophoto map 2931 CA 11 VERULAM

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions in the spaces provided in this question paper.
2. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
3. Indicate the unit of measurement or compass direction in the final answer of calculations, e.g. 10 km; 2,1 cm; west of true north.
4. You may use a non-programmable calculator.
5. The area demarcated in RED on the topographic map represents the area covered by the orthophoto map.
6. The following English terms and their Afrikaans translations are shown on the topographic map:

BACKGROUND INFORMATION ON VERULAM



Verulam is a town 28 kilometres north of Durban in KwaZulu-Natal, South Africa and is now part of the eThekweni Metropolitan Area. In 1850 a party of 400 Methodists settled here and formed the town. The town was then named after the Earl of Verulam, patron of the Brits who settled here. It is now a town inhabited mainly by Indians.

[Source: [https://en.wikipedia.org/wiki/Verulam,_KwaZulu](https://en.wikipedia.org/wiki/Verulam,_KwaZulu_Natal) Natal]

QUESTION 1 – MULTIPLE CHOICE AND CALCULATIONS

- 1.1 The questions below are based on the 1: 50 000 topographic map 2931 CA VERULAM as well as the orthophoto map 2931 11 VERULAM. Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) in the block next to each question. (4 x 1) (4)

- 1.1.1 The province located west of VERULAM is ...

- A. Gauteng.
- B. Limpopo.
- C. North West.
- D. Free State

D

- 1.1.2 The contour interval on the topographical map is ... metres.

- A. 5
- B. 10
- C. 20
- D. 25

C

- 1.1.3 The map index of the topographic map, south of 2931 CA VERULAM, is ...

- A. 2931 CC
- B. 2931 CD
- C. 2931 AC
- D. 2931 AA

A

- 1.1.4 The coordinates of Mount Moreland in block **D4** on the topographic map is ...

- A. 29°38'12" E 31°05'30" S
- B. 29°40'00" S 29°02'35" E
- C. 29°38'12" S 31°05'30" E
- D. 29°39'45" E 31°06'56" S

C

1.2 Refer to point G in block **B5** and point F in block **C5** on the topographic map.

- 1.2.1 Using the information on the topographic map, determine the magnetic bearing from point G in block **B5** to point F in block **C5** on the topographic map for the current year. Show ALL calculations. Mark will be awarded for calculations. (7 x 1) (7)

Formula: MB = True Bearing (TB) + Magnetic Declination (MD)

$$TB = 150^\circ$$

$$2024 - 2016 = 8 \text{ years}$$

$$8 \times 9' = 72'W$$

$$25^\circ 04' + 72' = 25^\circ 76' W \\ = 26^\circ 16'W$$

$$MB = 176^\circ 16' W$$

- 1.2.2 Why is it necessary to calculate the current magnetic declination? (1 x 2) (2)

Calculating the current magnetic declination is necessary for accurate navigation and orientation using a magnetic compass.

- 1.2.3 Refer to the trig beacon 70 in block **C3** and spot height 118 in block **B2** on the topographic map.

- (a) Calculate the average gradient between trig beacon 70 in block **C3** and spot height 118, in block **B2** on the topographic map.. (5 x 1) (5)

Formula: Average gradient = $\frac{\text{Vertical interval (VI)}}{\text{Horizontal equivalent (HE)}}$

$$VI = 124.7 - 118 = 6.7m$$

$$HE = 5.4 \text{ cm} \times 0.5 = 2.7km = 2700m$$

$$6,7 : 2700 \\ 1: 402,98$$

- (b) Is there intervisibility between trig beacon 70 in block **C3** and spot height 118 in block **B2**? (2 x 1) (2)

Yes or No: **Yes**

Give a reason for your answer: The hill that lies between the two points does not block or obscure line of sight between the two points.

QUESTION 2 - APPLICATION AND INTERPRETATION

2.1 The questions below are based on the 1: 50 000 topographic map 2931 CA VERULAM as well as the orthophoto map 2931 CA 11 VERULAM. Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) in the block next to each question.

2.1.1 The natural feature point J in block **D4** is a..... (1 x 1) (1)

- A. Windpump.
- B. Cultivated land.
- C. Non-perennial water.
- D. Reservoir.

C

2.1.2 The non-perennial river flowing in block **D4** on the topographic map is flowing in a ... direction.

- A. Western
- B. Southerly
- C. Northerly
- D. Easterly

D

(1 x 1) (1)

2.1.3 The Area calculation for the RED demarcated area of the orthophoto on the topographic map is ... in km² (1 x 1) (1)

- A. 1.5 KM
- B. 2.7 KM
- C. 2.1 KM
- D. 1.7 KM

B

2.2 Refer to settlements in block **D1** and in block **C5** on the topographic map.

2.2.1 Identify the settlement types in **blocks C4** and **D1** respectively. (2 x 1) (2)

C4: Rural

D1:Urban

2.2.2 The average temperatures of the settlement in block **D1** are higher than the settlement in block **C4**. What is this climatic phenomenon called? (1 x 1) (1)

Urban heat island

It 2.2.3 Why does settlement in block **D1** records higher temperatures? (1 x 2) (2)
It has more artificial surfaces that absorb heat.

2.2.4 What strategies can the municipality of Durban (eThekweni) implement to reduce the impact of this climate phenomenon (mentioned in question 2.2.2) (1 x 2) (2)

Greening and more reflective surfaces.

2.3 The western parts of Verulam experience valley winds at night.

2.3.1 Name the type of valley winds that affect the western parts of Verulam at night. (1 x 1) (1)

Katabatic winds

2.3.2 Katabatic winds are associated with temperature inversion. What is temperature inversion? (2 x 1) (2)

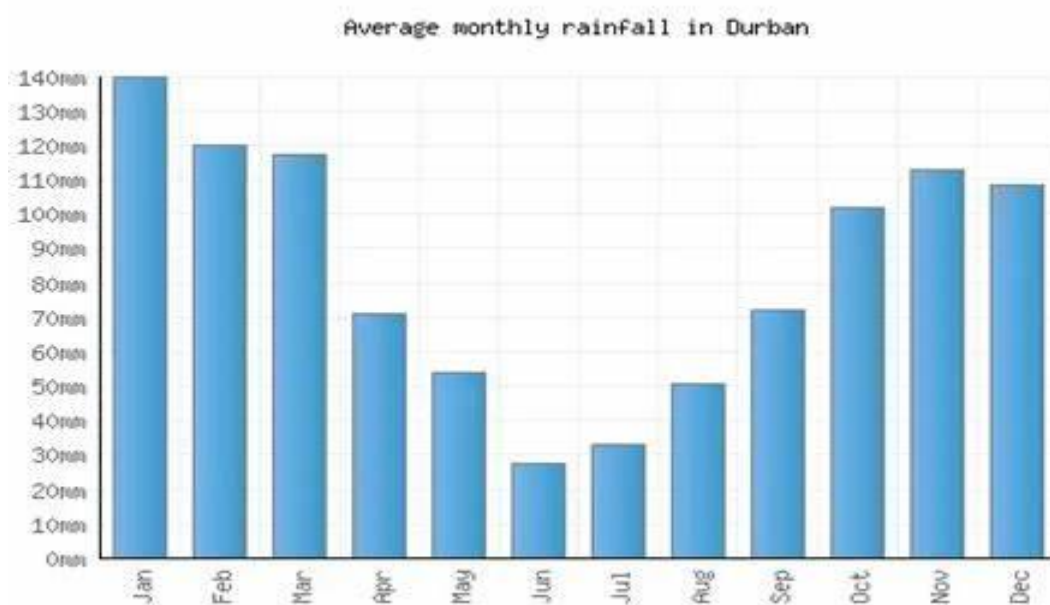
It is a reversal of normal conditions. The higher you go the warmer it gets.

2.3.3 Define temperature inversion?

(1 x 2) (2)

Give 2 marks

2.4 Refer to the graph below, showing the average monthly rainfall in Verulam.



2.4.1 The mapped area (on the topographic map) receives seasonal rainfall. Give two points of evidence from the topographic map to support this statement. (2 x 1) (2)

Presence of dams and non-perennial rivers

2.4.2 Provide the month during which Verulam received the lowest rainfall. (1 x 1) (1)
June

2.4.3 The northern part of Verulam is prone to flooding. Suggest measures evident on the topographic map (in block **B1**) to reduce the risk of flooding. (1 x 2) (2)

2.5 **Implement drainage systems in the area. Any reasonable answer.**
Refer to the passage and the picture below:



2.5.1 Due to pollutants shown above, Verulam experiences the effect of pollution dome. Define the concept pollution dome. (1 x 1) (1)

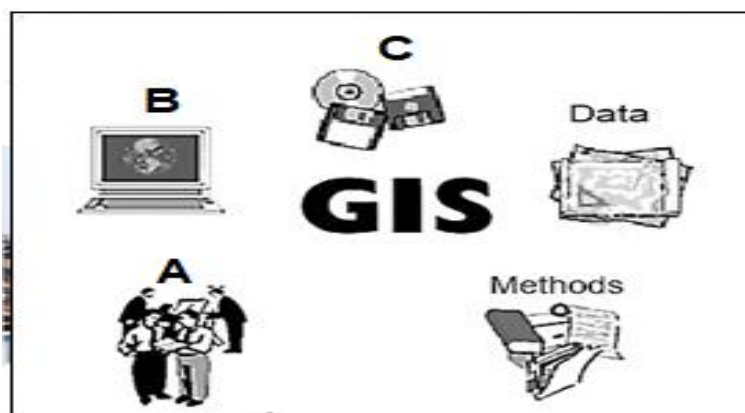
A mass of polluted air above a city/industrial area.

2.5.2 Mention TWO solutions that the local municipalities can implement to reduce the effect of pollution dome in Verulam. (2 x 2) (4)

**Plant more trees and move to more renewable sources of electricity generation.
Rapid bus systems and less cars on the road.**

QUESTION 3 – GEOGRAPHIC INFORMATION SYSTEMS

3.1 The following diagram shows the components of GIS.



3.1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A – D) in the box next to each question. (2 x 1) (2)

(a) Component **A** in the illustration above is...

- A software
- B hardware
- C people
- D processes

C

(b) Which of the following can be seen as a layer...

- A. Elevation
- B Temperature
- C Water
- D All of the above

D

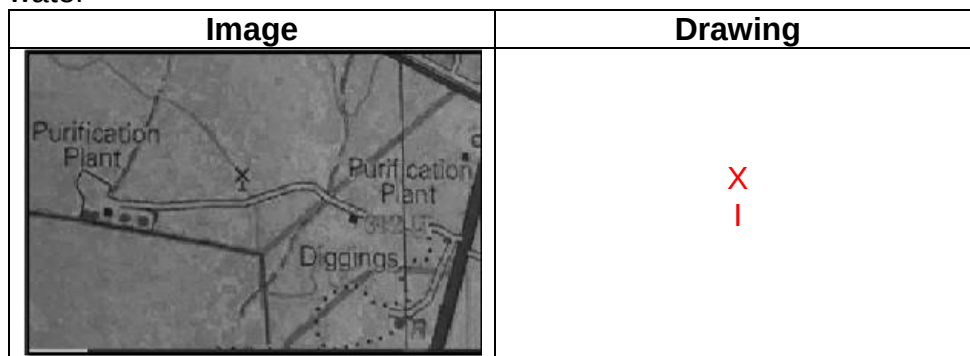
3.2 Refer to block **B4** on the topographic map.

3.2.1 Differentiate between the raster and vector data. (2 x 1) (2)
Raster is cell based, vector is co-ordinate based.

3.2.2 Identify a line feature in block **E1**. (1 x 1) (1)

Line : **A main road**

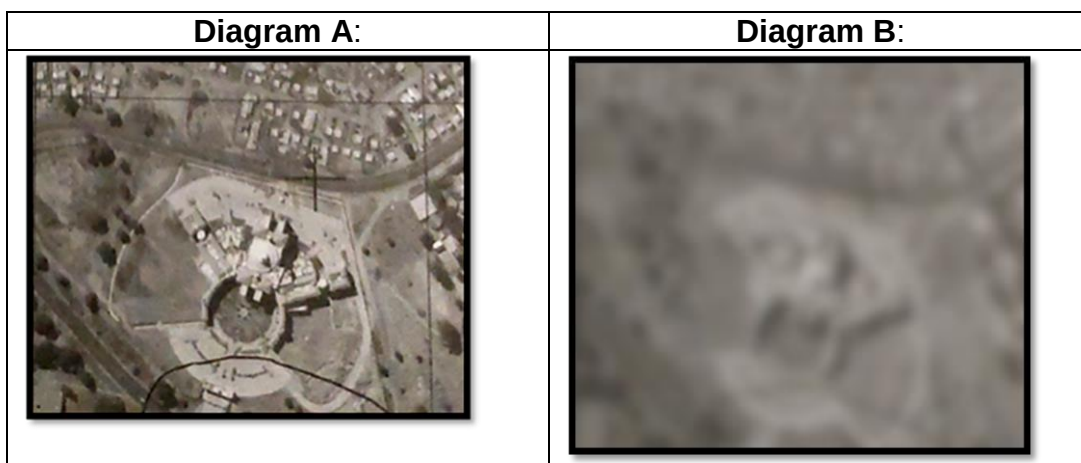
- 3.2.3 Refer to the image, of a portion of the topographic map, below and draw the symbol of the point feature that is used to extract ground water (1 x 1) (1)



- 3.2.4 What is the importance of the point feature identified in Question 3.2.3 to the cultivated land in this area? (1 x 2) (2)

An alternative water source.

- 3.3 Refer to diagram **A** and diagram **B** below.



- 3.3.1 Resolution refers to the detail to which a map depicts the location and shape of geographic features. Which image **A** or **B**, has a better resolution? (1 x 1) (1)

A

- 3.3.2 Give a reason for this difference in resolution between Image **A** and Image **B**. (1 x 2) (2)

More pixels

- 3.4 High traffic volumes and congestion on the main road 102 in block **D1** and **E1** on the topographical map map have been reported.

- 3.4.1 Identify the type road that joins the main road 102 in block **B4/D1** on the topographical map. (1 x 1) (1)

Any reasonable answer.

- 3.4.2 Suggest THREE ways in which the eThekweni municipality can use GIS to reduce traffic congestion in this area.

(3 x 1) (3)

(a) Analyse traffic patterns

(b) Identify bottlenecks

(c)

Traffic Flow Analysis:

- **Use of GIS:** Implement GIS to analyze and model traffic flow patterns in the municipality. GIS can incorporate data on road networks, traffic signals, and historical traffic patterns.
- **Application:** By understanding how traffic moves throughout the area, the municipality can identify congestion-prone areas and optimize signal timings, lane configurations, and road infrastructure to improve the overall flow of traffic. GIS can simulate scenarios to assess the impact of different traffic management strategies.

Optimized Routing and Navigation:

- **Use of GIS:** Utilize GIS to develop dynamic routing and navigation systems that consider real-time traffic conditions. GIS can integrate data on current traffic flow, road closures, and incidents.
- **Application:** Motorists can be provided with optimized routes in real-time, diverting them away from congested areas. This not only improves individual travel times but also helps distribute traffic more evenly across the road network, reducing congestion in specific locations.

Public Transportation Planning:

- **Use of GIS:** Employ GIS to plan and optimize public transportation routes, stops, and schedules. GIS can integrate data on population density, commuting patterns, and existing public transportation infrastructure.
- **Application:** By enhancing public transportation efficiency, the municipality can encourage more people to use public transit, reducing the number of private vehicles on the road. GIS can aid in identifying optimal locations for new transit hubs, bus routes, and park-and-ride facilities, promoting a more effective and attractive public transportation system.

Dave's law.

TOTAL : 60

ROUGH WORK